

EXAMPLE NAME: UART_EXAMPLE_XMC45

OVERVIEW: This example demonstrates the transmission and reception of messages from XMC4500 to the PC through UART interface.

DESCRIPTION:

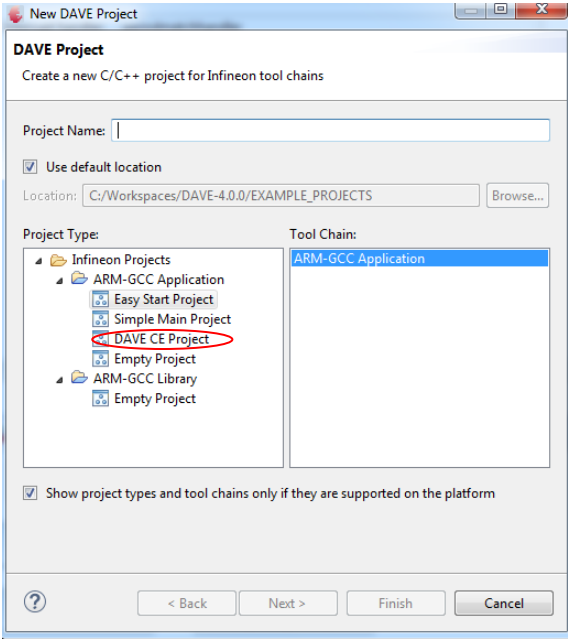
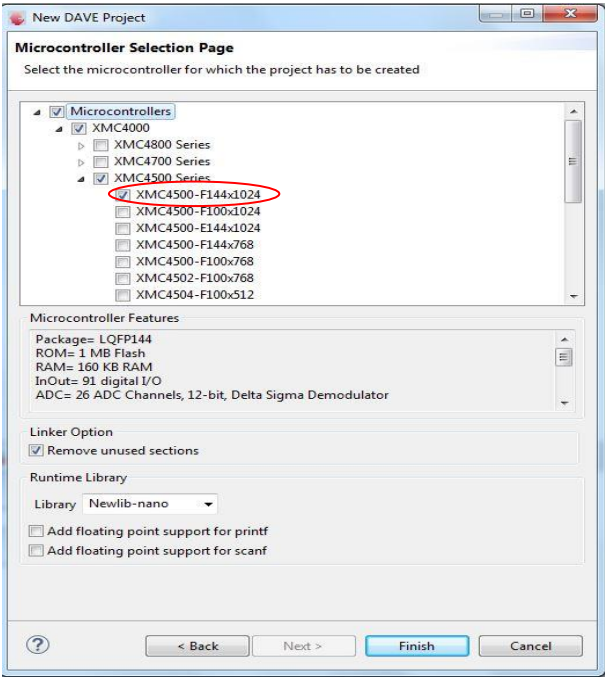
- UART APP Configures the USIC peripheral to send and receive messages to/from the PC using the UART Protocol. The example uses the transmit and receive pins connected to the on board debugger and its VCOM port in the host side.

REQUIRMENTS:

Boards Required: XMC4500 CPU Board

HOW TO CREATE THE PROJECT:

1. Open the DAVE CE and use  "Add IDE New Project Wizard" on the toolbar to add a new DAVE Project.

Enter Project Name Select "Project Type → DAVE CE Project"	Select Microcontroller → XMC4500 Series → XMC4500-F144X1024
	

- Use the  "Add New APP" in the toolbar to add 1 instance of UART. Configure the APP instance with the following configurations.

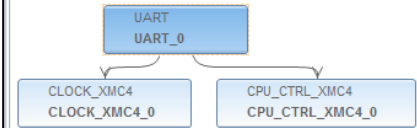
Views: APP Dependency Tree and APP Dependency

APP Dependency Tree

Search filter

- UART_0
 - CLOCK_XMC4_0
 - CPU_CTRL_XMC4_0

APP Dependency



UART:

General Settings | Advanced Settings | Interrupt Settings | Pin Settings

Operation mode: Full Duplex

Desired speed [baud]: 19200

Actual speed [baud]: 19200

Data bits: 8

Stop bit: 1 Stop Bit

Parity selection: No Parity

General Settings | Advanced Settings | Interrupt Settings | Pin Settings

Timing Settings

Oversampling: 16

FIFO Settings

☒ Enable transmit FIFO Size: 32

☒ Enable receive FIFO Size: 32

A total of 64 FIFO entries are available to be configured as transmit and receive buffers. These 64 entries are additionally shared between the two channels of a USIC module

- Use the  "Manual Pin Allocator" found in the toolbar, configure the pins for the UART.

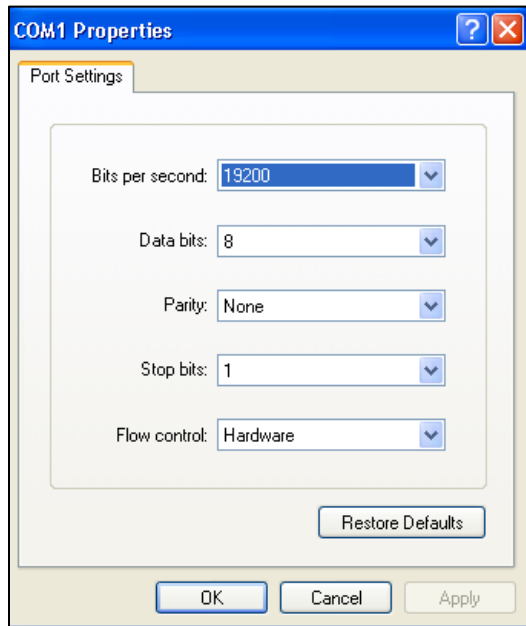
App Instance Name	App Pin Name	Pin Number (Port)
UART_0	Receive Pin	#108 (P1.4)
	Transmit Pin	#107 (P1.5)

- Generate the code for the configurations made and change main.c accordingly.
- Build and download to the microcontroller.

HOW TO TEST:

Look for the Jlink CDC UART Port number in the Device Manager of Windows TM

Start your favourite terminal tool on the PC with these settings for your respective COM port number:



Download and run the demo using the CPU_45A-V3 board. If everything is well configured, you will see "Press enter if you want to receive a message" in the terminal tool. Whenever you press "Enter", "Hello world!" will be displayed in the terminal tool.